

UNG FACILITIES PLANNING & GIS

Version 3.5

May 2026

Writes AD occupant data into GIS floor plan polygons across all five campuses

Campus_Floor_Plans_INFO_update_PROD_V3.5.py

Purpose

The core GIS update step. Reads the parsed AD file, applies both crosswalks, builds a building+room lookup table, then matches every GIS polygon against that table and writes three occupant fields. Creates a timestamped backup of each campus feature class before writing. Also exports unmatched records and — new in V3.5 — excluded records for diagnostic review.

When to Run

- Weekly, as Step 5 of the pipeline
- Always run the TEST version first to verify results before running PROD
- Run AFTER clearing fields (Step 4) and BEFORE the merge (Step 6)

Requirements

- ArcGIS Pro / arcpy
- Parsed AD Excel from Step 1
- Building_Crosswalk.xlsx
- Department_to_Domain_CrosswalkV2.xlsx

Inputs

Item	Value	Notes
Parsed AD Excel	FacStaffWithAddress_MMDDYY YY_parsed.xlsx	Output of Step 1
Building Crosswalk	Building_Crosswalk.xlsx	95 verified AD name → GIS code mappings
Department Crosswalk	Department_to_Domain_CrosswalkV2.xlsx	108 department → COL_INS_CTR code mappings
5 PROD Feature Classes	UNG_ENTERPRISE_UPDATE D_WORKING.gdb	BRC, CC, DC, GC, OC FLOORPLANS

Three-Step Process

Step 1 — Build Lookup Table: Applies building and department crosswalks; filters suspect records; applies DEHYPHEN_BUILDINGS normalization; groups by building + room to handle multi-occupant rooms

Step 2 — Analysis: Compares lookup table against GIS polygons; identifies matches, duplicates to skip, and unmatched records; exports diagnostic files; requires y/n confirmation before proceeding

Step 3 — Update: Creates timestamped backup of each FC; writes OFF_ASGNE, asgn_to, and SPACE_INFO for matched polygons

Outputs

Item	Value	Notes
Updated feature classes	UNG_ENTERPRISE_UPDATE D_WORKING.gdb	1,150 polygons populated across 5 campuses (May 21, 2026 run)
Backup feature classes	Same GDB	One per campus: e.g. BRC_FLOORPLANS_backup_Y YYYYMMDD_HHMMSS
Unmatched_AD_Rooms_TIMES TAMP.xlsx	ActiveDirectoryParsing\	Records in lookup not matched to a polygon; includes Reason column (no_match / duplicate)
Excluded_AD_Records_TIMES TAMP.xlsx	ActiveDirectoryParsing\	Records dropped before lookup was built; includes Reason (no_gis_bldg / no_room / no_surname)

DEHYPHEN_BUILDINGS

Some buildings use hyphenated sub-room suffixes in AD (e.g. 303-A) but GIS polygons do not (303A). The DEHYPHEN_BUILDINGS set currently contains DA008A (Hansford Hall). Hyphens are stripped from RoomClean at lookup-table build time and from GIS room values at cursor read time — GIS polygons are never modified. When AD entries for DA008A are eventually cleaned up, remove DA008A from this set in both PROD and TEST scripts.

Version Notes

V3.5: Added Excluded_AD_Records diagnostic output with Reason column (no_gis_bldg, no_room, no_surname)

V3.4: Added Reason column to Unmatched output; duplicate rows now included in unmatched file; fixed banner version string

V3.3: Added DEHYPHEN_BUILDINGS and normalize_room; applied at all four cursor locations and in prepare_lookup_table